Bt512 midterm mcqs

microbial biology (Virtual University of Pakistan)

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Section 7:Industrial and Environmental Microblology

1. Which microorganism is commonly used in the production of riboflavin (vitamin B2)?

- *A) Clostridium acetobutylicum*
- *B)Ashbya gossypii*
- *C) Saccharomyces cerevisiae*
- D)E.coli

Answer:B

2. The production of xanthan gum is associated with:

- A) *Streptococcus*
- B) *Xanthomonas campestris*
- C) *Aspergillus niger*
- D) *Lactobacillus*

Answer:B

3. Which microbe is used for the production of bioethanol fromlignocellulosic biomass?

- *A) Pseudomonas*
- *B) Clostridium thermocellum*
- *C) Saccharomyces cerevisiae*
- D)E.coli

Answer:C

4. Which process is most efficient in reducing greenhouse gases using microbes?

- A)Composting
- B)Methanogenesis
- C) Biofixation of CO2
- D) Nitrification

Answer:C

5. Biohydrogen production is carried out by which type of bacteria?

- *A)Cyanobacteria*
- *B)Methanogens*
- C) *Nitrobacter*
- D) *Lactobacillus*

Answer:A

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Section 8:Microbial **Genetic Engineering**

6. What is the primary advantage of CRISPR-Cas9 in genetic engineering?

- A)High mutation rate
- B) Target-specific gene editing
- C) Production of recombinant proteins
- D)Amplification of DNA

Answer:B

7. Ti plasmid, commonly used in plant genetic engineering,is derived from:

- *A)Escherichia coli*
- *B)Agrobacterium tumefaciens*
- *C) Bacillus thuringiensis*
- D) *Rhizobium*

Answer:B

8. Which enzyme is essential for joining DNA fragments during cloning?

- A) DNA helicase
- B) DNA polymerase
- C) DNA ligase
- D)Topoisomerase

Answer:C

9. A bacterium modified to produce human insulin is an example of:

- A)Gene therapy
- B) Transgenic microorganism
- C) Mutant strain
- D) Natural selection

Answer:B

10. What role does a selectable marker play in genetic cloning?

- A) To bind the plasmid to the DNA
- B) To identify transformed cells
- C) To amplify DNA sequences
- D) To replicate the host cell

Answer:B

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Section 9:Applications In Medical Blotechnology

11. What is the main microbial source of the antibiotic erythromycin?

- *A) Penicillium chrysogenum*
- B) *Streptomyces erythraeus*
- C) *Bacillus subtilis*
- D) *Lactobacillus acidophilus*

Answer:B

12. The production of monoclonal antibodies involves which cell fusion technique?

- A) Gene cloning
- B) Hybridoma technology
- C) CRISPR-Cas9
- D)Biotransformation

Answer:B

13. Interferons produced by recombinant bacteria are used for:

- A)Treating bacterial infections
- B) Combating viral diseases
- C) Enhancing food shelf life
- D) Producing vaccines

Answer:B

14. Which of the following is a microbial vaccine?

- A) Polio vaccine
- B) BCG vaccine
- C) Influenza vaccine
- D) Both B and C

Answer:D

15. The first microbial enzyme approved for medical use was:

- A)Amylase
- B) Penicillinase
- C) Streptokinase
- D)Lactase

Answer:C

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Section 10:Advances In Microblal Blotechnology

16. Which of the following is used in microbial biosensors?

- A)E.coli
- *B) Pseudomonas putida*
- C) *Saccharomyces cerevisiae*
- D) Both A and B

Answer:D

17. The production of single-cell protein (SCP) involves the use of:

- A)Algae
- B)Yeast
- C)Fungi
- D) All of the above

Answer:D

18. The term 'metagenomics' refers to:

- A) The study of genomic sequences
- B) The collective study of microbial genomes in an environment
- C) Single-gene analysis
- D) None of the above

Answer:B

19. Which microbial species is used for the production of statins?

- *A) Penicillium citrinum*
- B) *Aspergillus oryzae*
- C) *Saccharomyces cerevisiae*
- D) *Bacillus thuringiensis*

Answer:A

20. Which advanced technology is used for microbial genome editing?

- A)CRISPR-Cas9
- B) RNA interference
- C)Plasmid transformation
- D)Electroporation

Answer:A

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